Polymers

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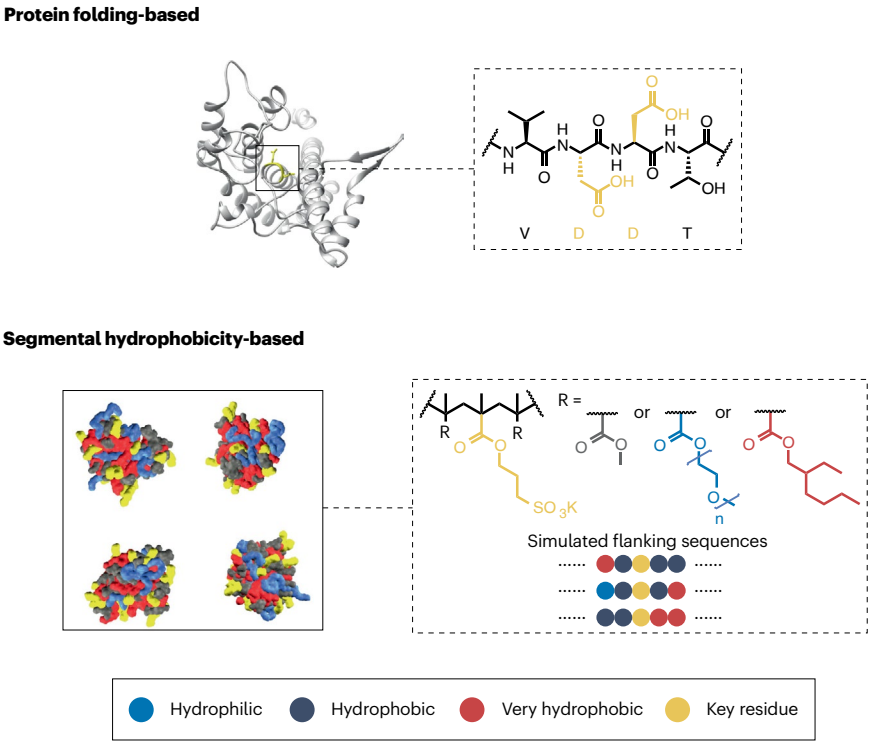
Mimicking enzymes

The interpretation of the sequence–structure–function relationship in folded proteins represents a challenge for chemists and material scientists. At the same time, there are common traits that can have an overall influence: Now, writing in *Nature*, Alexander-Katz, Xu and co-workers analyse 1,300 metalloproteins to understand and rationalize these traits: indeed, they find that the hydrophobicity of certain residues can in fact modulate the spatial distribution and dynamics of sidechains, while cofactor binding at specific residues can affect the rearrangement of such side-chains and further tailor the reactive environment.

Based on their findings, the group of researchers designed random heteropolymers (RHPs) that mimic the activity of folded proteins. The group's basic RHP-P1 contained methyl methacrylate (MMA), 2-ethylhexyl methacrylate (2-EHMA), oligo(ethylene glycol) methyl ether methacrylate (OEGMA, M_w ~ 500) and 3-sulfopropyl methacrylate potassium salt (SPMA). Increasing the sulfonic acid-containing but hydrophobic SPMA while reducing the OEGMA fraction resulted in RHP-S1 – which mimics the catalytic site of squalene hopene cyclases where a polar aspartic acid is a key catalytic residue surrounded by a non-polar environment in the cyclization of citronellal (CA). The researchers then studied and characterized how the monomer composition impacted the structures and dynamics of these polymers, using structural techniques and computational simulations. Although the two polymers have similar segmental hydrophobicity – which is comparable to that of proteins – RHP-S1 provided a more hydrophobic environment than RHP-P1 beside the key monomer SPMA, resulting in higher activity for the cyclization of CA.



An additional design involved the introduction of histamine-modified methacrylic acid *N*-hydroxy succinimide ester, which mirrored the haeme binding histidine in peroxidases, resulting in RHP-H1. From UV–Vis and electron paramagnetic resonance studies, the group of researchers confirmed that RHP-H1 binds haeme in a 1:2 RHP/haeme ratio, independently of the polymer composition, and this polymer has peroxidase activity, as confirmed by assays with H_2O_2 and 2,2′-azino-di(3-ethylbenzthiazoline-6-sulfonic acid) and with the degradation of tetracycline. Further designs of these enzyme mimics were also adapted for electrospinning of mats with tetracycline degrading activity.

The modulation of polymer chains and active residues demonstrated in the researchers' study can indeed replicate proteins' behaviour. It can also be adapted to materials that can retain such catalytic activity with an expanded scope under non-biological conditions, resulting in enzyme-like materials for application in environmental remediation and the plastics industry.

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